

Causal Inference

Lecture 7

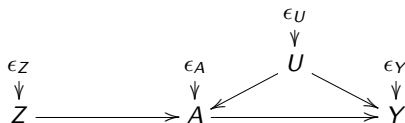
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Structural equations (SEs)

- A SE describes the causal mechanism that generates the outcome variable.
- For the DAG above, the SEs are

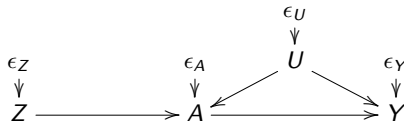
$$Z = f_Z(\epsilon_Z),$$

$$U = f_U(\epsilon_U)$$

$$A = f_A(Z, U, \epsilon_A)$$

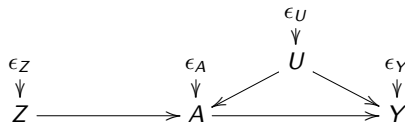
$$Y = f_Y(A, U, \epsilon_Y)$$

1. The structural equations contain only direct (causal) effects.
2. Usually, the noise variables ϵ s are implicit in the DAGs. However, appear explicitly in the SEs to denote the non-deterministic generation of the outcome.
3. The equality in SEs is asymmetric (that is, the equality is an abuse of notation. The formal notation is $:=$). Namely, if one changes the subject of a structural equation, it loses its structural (causal) interpretation.



Endogenous and exogenous variables

- The variables generated by other variables in the DAG are called endogenous. In the DAG above, A and Y are endogenous.
- Variables generated “outside” the DAG, namely, have no parent variables in the DAG, are called exogenous. In the DAG above, $\{Z, U, \epsilon_Z, \epsilon_U, \epsilon_A, \epsilon_Y\}$ are exogenous.
- Although the structural equations above are non-parametric (the f_Y may not follow a parametric structure), we usually specify a parametric distribution for either the distribution of the noise term or its mean.



Parametric and Semi-parametric Structural Equations

- For example, for the last SE for Y , if we denote

$$\begin{aligned} Y &= f(A, U; \beta) + \epsilon_Y \\ &= \beta_0 + \beta_A A + \beta_U U + \epsilon_Y, \end{aligned}$$

where $\epsilon_Y \sim N(0, \sigma^2)$, then the model is fully parametric.

- However, if we specify only its mean

$$\begin{aligned} \mathbb{E}[Y|A, U] &= f_Y(A, U; \beta) \\ &= \beta_0 + \beta_A A + \beta_U U, \end{aligned}$$

then it is semi-parametric since we remain agnostic regarding the distribution of ϵ_Y , which may be non-parametric. We assume only that it has mean zero and finite variance.

Structural mean models

- Let Y be continuous outcome, A binary treatment, and L are measured confounder that assumed to be sufficient for confounding control. Namely, we have conditional exchaingibility

$$Y_a \perp\!\!\!\perp A | L$$

- Assume that the target parameter is the average treatment effect (ATE), i.e.,

$$\mathbb{E}[Y_1 - Y_0]$$

- We specify the structural mean model for the potential outcome

$$\mathbb{E}[Y_a | L] = \beta_0 + \beta_1 a + \beta_2 L + \beta_3 aL,$$

- Using the conditional exchangeability, this is also the outcome model, i.e., as $\mathbb{E}[Y_a | L] = \mathbb{E}[Y_a | A = a, L] = \mathbb{E}[Y | A = a, L]$.
- Notably, the $\mathbb{E}[Y_0 | L]$ is nested in the structural model for the treated, $\mathbb{E}[Y_a | L]$

$$\mathbb{E}[Y_0 | L] = \mathbb{E}[Y | A = 0, L] = \beta_0 + \beta_2 L,$$

thus,

$$\mathbb{E}[Y_1 | L] = \mathbb{E}[Y_0 | L] + \beta_1 + \beta_3 L.$$

- Although we specify the complete structural mean model

$$\mathbb{E}[Y_a|L] = \beta_0 + \beta_1 a + \beta_2 L + \beta_3 aL,$$

we are interested in the ATE.

- The conditional ATE can be written as

$$\mathbb{E}[Y_1 - Y_0|L] = \beta_1 + \beta_3 L,$$

- Therefore, the ATE is

$$\mathbb{E}[Y_1 - Y_0] = \beta_1 + \beta_3 \mathbb{E}[L].$$

- If we are using the standardization (G-computation) method we need to estimate also the parameters of the $\mathbb{E}[Y_0|L]$ model, i.e., β_0 , and β_2 . However, these are nuisance parameters that we are not interested in.

- A semiparametric structural mean model (SMM) only specifies a model for the causal contrast on interest (e.g., ATE)

$$\mathbb{E}[Y_1 - Y_0|L] = \beta_1 + \beta_3 L.$$

- This model is semiparametric since we specify a model for the mean and not the distribution of the outcome, and, more important, we are agnostic regarding the structure of $\mathbb{E}[Y_0|L]$, which is allowed to be non-parametric.
- Therefore, the SMM is

$$\mathbb{E}[Y(1) - Y(0)] = \beta_1 + \beta_3 \mathbb{E}[L].$$

- That can be estimated by estimating β_1 and β_3 , and then marginalize over L , i.e.,

$$\hat{\mathbb{E}}[Y(1) - Y(0)] = \hat{\beta}_1 + \hat{\beta}_3 \sum_{i=1}^n L_i.$$

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Homogeneous causal effect

- To simplify the introduction, we assume a homogeneous causal effect ψ for all $i = 1, \dots, n$, i.e.,

$$Y_{1,i} - Y_{0,i} = \psi,$$

- This can be rewritten as

$$Y_{0,i} = Y_{1,i} - \psi,$$

namely, the potential outcome for $a = 0$, $Y_{0,i}$, equals the potential outcome for $a = 1$ minus the causal effect of the treatment.

- Assume that the i th subject was in-fact treated, hence, $Y_{1,i}$ is the observed outcome.
- Therefore, using the consistency assumption, we have $Y_{1,i} = Y_i$, hence,

$$Y_{0,i} = Y_i - \psi.$$

- Namely, if we know ψ , we can predict the counterfactual Y_0 for the i th subject.

Conditional exchangeability

- We define $H(\psi)$ as follows

$$h_i(\psi) = Y_i - \psi A_i.$$

- For the treated subjects, $A_i = 1$, $h_i(\psi) = Y_i - \psi$ is the predicted counterfactual $Y_{0,i}$, while for the untreated, $h_i(\psi) = Y_i$, is the observed outcome Y .
- Therefore, the goal is to find the right value of ψ that is the causal effect of the treatment. Namely, the value that accurately predicts the counterfactual value $Y_{0,i}$.
- By conditional exchangeability, $Y_0 \perp\!\!\!\perp A|L$. Therefore, we can find the value of ψ that makes the predicted counterfactual (conditionally) independent of the treatment assignment, i.e.,

$$h(\psi) \perp\!\!\!\perp A|L.$$

Non-conditional exchangeability

- Assume for clarity of illustration that $L = \emptyset$, thus, we have non-conditional exchangeability $h(\psi) \perp\!\!\!\perp A$

$$\sum_{i=1}^n h(\hat{\psi}) A_i = \sum (Y_i - \hat{\psi} A_i) A_i = 0$$

therefore, solving for ψ , we have

$$\hat{\psi} = \frac{\sum Y_i A_i}{\sum A_i^2}.$$

- Which is the mean value for the treated, i.e., $\hat{\psi} = \bar{Y}$ for $A = 1$, which is also the estimator of the ATE.

Conditional exchangeability

- Assume now that $L \neq \emptyset$, thus, we have conditional exchangeability $h(\psi) \perp\!\!\!\perp A|L$, hence the estimating equation is

$$\sum_{i=1}^n h(\hat{\psi})(A_i - \mathbb{E}[A|L_i]) = \sum (Y_i - \hat{\psi}A_i)(A_i - \mathbb{E}[A|L_i]) = 0$$

therefore, solving for ψ , we have

$$\hat{\psi} = \frac{\sum Y_i(A_i - \mathbb{E}[A|L_i])}{\sum A_i(A_i - \mathbb{E}[A|L_i])},$$

which is the estimator of the ATE.

- Note that if we condition on some continuous L , we, practically, need to fit a model for the exposure, i.e., $\mathbb{E}[A|L]$.
- The main idea of the G-estimation is finding the value of ψ that makes the counterfactual outcome Y_0 independent of the treatment assignment A in the given sample.**
- Practically, it means equating to zero the scalar product of $h(\psi)$ and $A - \mathbb{E}[A|L]$.

- A generalized structural mean model (GSMM) is

$$\xi(\mathbb{E}[Y_a|L, A = a]) - \xi(\mathbb{E}[Y_0|L, A = a]) = m_a^T(L)\psi_a,$$

where ξ is a link function of a generalized linear model, ψ_a is the vector of causal parameters, and $\dim(m_a(L)) = \dim(\psi_a)$. The composition of $m_a(L)$ defines the exact form of the causal model.

- The general form of the $H(\psi)$ function is

$$h(\psi) = \xi \left(\xi^{-1}(\mathbb{E}[Y|L, A = a]) - m^T(L)A\psi \right)$$

where $\mathbb{E}[Y|L, A = a]$ reflects an intermediate step of modeling the conditional expectation of the observed outcome Y .

Examples of link functions

$$h(\psi) = \begin{cases} Y - m^T(L)A\psi, & \text{identity} \\ Y \exp\{-m^T(L)A\psi\}, & \text{log} \\ \text{expit}(\text{logit}(\{\mathbb{E}[Y|L, A = a]\}) - m^T(L)A\psi), & \text{logit} \end{cases}$$

Two parameters model

- Assume the following SMM

$$\mathbb{E}[Y_a|A = a, L] - \mathbb{E}[Y_0|A = a, L] = \psi_0 a + \psi_1 aL,$$

namely, there are two causal parameters ψ_0 and ψ_1 .

- The predictor of the counterfactual Y_0 is

$$h_i(\psi) = Y_i - \psi_0 A_i - \psi_1 A_i L_i.$$

- The number of estimating equations equals the number of unknown parameters, therefore, we should have two estimating equations.

For ψ_0 we can choose

$$\sum h_i(\psi)(A_i - \mathbb{E}[A|L_i]) = 0.$$

For ψ_1 we can choose

$$\sum h_i(\psi)L_i(A_i - \mathbb{E}[A|L_i]) = 0.$$

- The values of $\psi^T = (\psi_0, \psi_1)$ that solve these two equations simultaneously, are the G-estimators.

General form

- For some $h(\psi)$, and a function d , the estimating equations are given by

$$\sum h_i(\psi)(d(A_i, L_i) - d(\mathbb{E}[A|L_i], L_i)) = 0.$$

- The values of ψ^T that solve these two equations simultaneously, are the G-estimators.
- The function d is arbitrary, as long as the equations are linearly independent and

$$\mathbb{E}[d(A_i, L_i) - d(\mathbb{E}[A|L_i], L_i)|L] = 0.$$

- The most common choice (and, for a case of constant variance, also the best choice) is

$$m(L)(A - \mathbb{E}[A|L]),$$

where $m(L)$ can also be derived as the gradient of $h(\psi)$ w.r.t ψ , i.e.,

$$m(L) = \frac{\partial}{\partial \psi} h(\psi).$$

- In order to use G-estimation we need two models and an estimating equation

1. The semiparametric SMM

$$\mathbb{E}[Y_1 - Y_0 | A = 1, L] = m^T(L)\psi,$$

where $m(L)$ may allow for interaction terms between the treatment effect and the covariates L . The dimension of ψ is the same as $m(L)$.

2. Exposure/treatment model

$$\mathbb{E}[A|L] = \alpha^T L.$$

3. Solution w.r.t. ψ of the set of estimating equations

$$\sum m(L_i)(Y_i - m^T(L_i)\psi)(A_i - \alpha^T L_i) = 0.$$

Example

- Suppose that the SMM is

$$\mathbb{E}[Y_1 - Y_0|L] = \psi_0 + \psi_1 L$$

- The corresponding $h(\psi)$ is

$$h_i(\psi) = Y_i - \psi_0 A_i - \psi_1 A_i L_i$$

- The treatment model is

$$\mathbb{E}[A|L] = \alpha_0 + \alpha_1 L,$$

where α_0, α_1 are unknown parameters that can be estimated using the classical score equations.

- The G-estimator of ψ are the values that satisfy the conditional exchangeability in the given sample

$$h(\psi) \perp\!\!\!\perp A|L.$$

Therefore, the set of estimating equations is

$$\sum_{i=1}^n \begin{pmatrix} 1 \\ L_i \end{pmatrix} \begin{pmatrix} Y_i - \psi_0 A_i - \psi_1 A_i L_i \\ Y_i - \psi_0 A_i - \psi_1 A_i L_i \end{pmatrix} \begin{pmatrix} A_i - \hat{\alpha}_0 - \hat{\alpha}_1 L_i \\ A_i - \hat{\alpha}_0 - \hat{\alpha}_1 L_i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Three parameters model example

1. Assume the following SMM

$$\mathbb{E}[Y_a|A = a, L] - \mathbb{E}[Y_0|A = a, L] = \psi_0 a + \psi_1 aL + \psi_2 aL^2,$$

namely, there are three causal parameters ψ_0 , ψ_1 , and ψ_2 .

2. The predictor of the counterfactual Y_0 is

$$h_i(\psi) = Y_i - \psi_0 A_i - \psi_1 A_i L_i - \psi_2 A_i L_i^2.$$

3. The number of estimating equations equals the number of unknown parameters, therefore, we should have three estimating equations.

$$\sum h_i(\psi) \begin{pmatrix} 1 \\ L_i \\ L_i^2 \end{pmatrix} (A_i - \mathbb{E}[A|L_i]) = 0.$$

4. The values of $\psi^T = (\psi_0, \psi_1, \psi_2)$ that solve these two equations simultaneously, are the G-estimators.

Algorithm

1. To use the G-estimation, we need to specify a semiparametric structural mean model for the (conditional) causal contrast

$$\xi(\mathbb{E}[Y_a|A=a, L]) - \xi(\mathbb{E}[Y_0|A=a, L]) = m^T(L)\psi$$

2. We specify $h(\psi)$ that satisfy $\mathbb{E}[h(\psi)|L] = \mathbb{E}[Y_0|L]$.
3. We fit a model for the exposure/treatment, i.e., $\mathbb{E}[A|L]$.
4. Construct the estimating equations $\sum h_i(\psi)m(L)(A_i - \mathbb{E}[A|L_i]) = 0$.
 - The values of ψ^T that solve these two equations simultaneously, are the G-estimators.
 - The variance of the G-estimators is computed using the sandwich formula (next chapter).
 - G-estimation is an alternative to standardization and IPW for estimating causal parameters. Its key advantage is that it avoids strict parametric assumptions about $\mathbb{E}[Y_0 | L]$, thereby allowing more flexible models of the untreated potential outcome.
 - The full potential of G-estimation methods becomes particularly evident when instrumental variables are introduced, showcasing their utility in complex causal inference settings.

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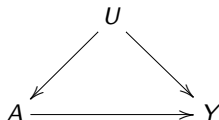
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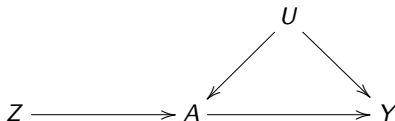
The examined problem:



- Simple causal structure where U confounds the effect of A on Y .
- Our goal is to identify and estimate the causal effect of A on Y .
- We assume that U is unmeasured (unobserved).

Identification and Estimation of the causal effect $A \rightarrow Y$: using IV

- When U is unmeasured, one can still identify the causal effect of interest by using a special variable Z .
- Such a variable is associated monotonically with the exposure A (relevance), affects the outcome Y only through the exposure (exclusion), and is not confounded with the outcome (exogeneity).
- These three properties define a valid instrumental variable (IV). Such variables allow for the identification of causal effects in the presence of unmeasured confounders U .



DAG of a valid instrumental variable Z

The aim in most epidemiological and medical studies

- Estimate the causal effect of the exposure on the outcome.

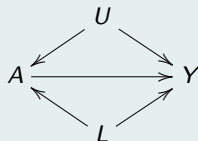
Problems to overcome

1. The optimal setting for causal inference is the conduction of randomised control trials (RCT). However, most of the epidemiological data is observational.
2. Possible unmeasured confounding variables may jeopardise the estimators' validity (inconsistency).
3. The traditional estimation method deals efficiently only with linear models, whereas most common models in epidemiological, medical and econometrical research are non-linear.

Causal effect estimation and the `ivtools` R-package

- Estimate the causal effect of the exposure on the outcome in the presence of unmeasured confoundings by
 - ➊ Using instrumental variable (IV) methods - overcomes (under suitable conditions) the confounding bias.
 - ➋ Using *G*-estimation - allows for using the IV in non-linear models.
 - ➌ Perform the analysis in linear and non-linear models (GLMs) with the `ivtools` R-package.

Dealing with confounders in observational studies



- In observational studies, we usually don't have exchangeability because we cannot ensure that the measured covariates are sufficient for confounding control.
- In principle, we may achieve conditional exchangeability by controlling for an appropriate set of covariates:

$$(Y_0, Y_1) \perp\!\!\!\perp A \mid U, L.$$

- In practice, the measured covariates L are insufficient for confounding control.

Basic notations

- Exposure: A , numerical (binary or continuous)
- Outcome: Y , numerical, measured at one specific time point.
- Instrumental variable: Z , numerical (scalar or vector).
- Measured covariates: L , scalar or vector (or an empty set). Possible confounders for A and Y
- Unmeasured confounders: U .

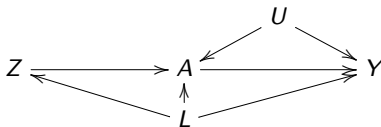
The need for IV

- When exposure is not randomly assigned to participants, the causal effect of the exposure cannot be recovered from simple regression methods.
- A special type of variable called Instrumental can be used to recover the causal effect of the exposure on the outcome even in non-randomised settings.
- IV methods is a collection of methods that can be used to estimate causal effects in the presence of unmeasured confounders U .

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Instrumental variables

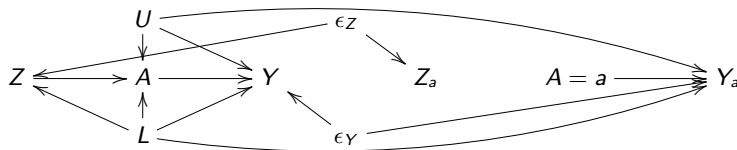
- A possible approach to address the issue of confounding bias is to use instrumental variables (IVs). An IV Z is a variable associated with exposure A , is not associated with unmeasured confounders (exogeneity), and affects the outcome Y only through the exposure A (exclusion).
- Let $\{L, U\}$ be a set of all confounders of the exposure A and the outcome Y , such that L denotes measured confounders, and U the unmeasured confounders.



A causal structure of a valid IV. The instrument Z affects Y only through the exposure A , and is not confounded with the outcome by the unmeasured variables.

Twin-causal network

- The exclusion and the exogeneity assumptions can be compactly formulated using counterfactuals: $Y_a \perp\!\!\!\perp Z|L$, $a \in \{0, 1\}$. Such a formulation avoids explicit reference to U .
- Directed acyclic graph (DAG) of a twin network causal model with instrumental variable. The left-hand side of the DAG represents the observed actual world, while the right-hand side represents the hypothetical potential world.
- Since Z is not affected by the exposure, $Z_a = Z$.
- For both DAGs, ϵ_Z and ϵ_Y represent all the unmeasured exogenous factors that determine the values of Z and Y , respectively.



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A causal model

- Let Y_0 be the potential outcome for a given subject with $A = 0$.
- Let $m(L)$ be a vector function of the measured variables L .
- Let Z be the instrumental variable.
- Let ξ be the link-function.
- The implicit causal model is given by

$$\xi(\mathbb{E}[Y|L, A, Z, U]) = m^T(L)A\psi + \gamma L + f(U)$$

where the parameter ψ measures the causal effect of the particular exposure level $A = a$ (among those who actually received the exposure level a), conditional on L , Z , and U .

- For identifiability, we assume that the causal effect is constant across instrument levels Z .

Generalized structural mean model

- The generalized structural mean models (GSMM) are semiparametric two-stage models that can handle, among other settings, binary outcomes.
- The first stage model is the generalised structural model indexed with causal parameters that convey the causal effects of interest.
- The second stage model is the association generalised model for the observed data that is required to identify the causal parameters of the former.
- The structural and association models' two link functions can differ. However, possible issues are introduced in such a case when there are no unmeasured confounders.
- Notably, as the association model models the observed data, all its features can be empirically tested. At the same time, the structural model is a latent model defined using counterfactuals. Thus, its features cannot be empirically tested without additional restricting assumptions.

Generalized structural mean model

- A generalised structural mean model (GSMM) is

$$\xi(\mathbb{E}[Y_1|Z, L, A = a]) - \xi(\mathbb{E}[Y_0|Z, L, A = a]) = m_a^T(L)\psi_a,$$

where ξ is a link function of a generalized linear model, ψ_x is the vector of causal parameters, and $\dim(m_a(L)) = \dim(\psi_a)$. The composition of $m_a(L)$ defines the exact form of the causal model.

- Assume that the association model of the observed outcome follows a parametric structure indexed by the vector of coefficients β_Y , i.e., $\xi(\mathbb{E}[Y|Z, L, A; \beta_Y])$. Such models are referred to as double ξ models.
- Although the unmeasured confounders are implicit, the consequence of their omission is incorporated in the values of β_Y . Specifically, controlling for the exposure A in the setting with unmeasured confounders U opens a non-causal path between Z and the observed outcome Y and the potential outcome Y_a .
- Formally, A is a collider, thus, conditioning on A opens the non-causal paths

$$Z \rightarrow A \leftarrow U \rightarrow Y \quad \text{and} \quad Z \rightarrow A \leftarrow U \rightarrow Y_a,$$

therefore, Z appears in the GSMM above.

Omitting U from the Latent Causal Model Formulation

- A generalized structural mean model (GSMM) can be written as:

$$\xi(\mathbb{E}[Y_1 \mid Z, L, A = a]) - \xi(\mathbb{E}[Y_0 \mid Z, L, A = a]) = m_a^T(L) \psi_a,$$

where $\xi(\cdot)$ is a link function (as in a generalized linear model), ψ_a is the vector of causal parameters, and $\dim(m_a(L)) = \dim(\psi_a)$. The choice of $m_a(L)$ determines the exact form of the causal model.

- Here, Z is **not** merely conditioned upon in the usual sense; rather, it serves as a valid instrument that enables identification of the causal parameters ψ_a .
- Even if unmeasured confounding U exists (or is not directly modeled), a valid instrument Z can still identify the causal effect, thus avoiding the need to condition explicitly on U .
- The use of Z is **not** simply regressing Y on Z ; rather, one applies suitable IV-based estimators such as two-stage least squares (2SLS) or G-estimation to consistently estimate the causal parameters.

Causal effect in linear model

Assume that ξ is the identity link function, and that the exposure A is binary, then

$$\mathbb{E}[Y|A = 1, L, Z, U] - \mathbb{E}[Y_0|A = 1, L, Z, U] = m(L)^T \psi.$$

- If $m(L) = 1$ then the exposure effect is just the parameter of interest itself ψ , which represents the effect of increasing the exposure level A from 0 to 1, given L , Z and U , and $A = 1$.
- In the general case of $m(L)^T A \psi$, the vector valued function $m(L)$ allows for interaction between the exposure A and the measured variables L .
- To allow for the main causal effect of the exposure A on the outcome Y , the first element of $m(L)$ is typically set to 1.
- Note that once we condition on U , Z is independent (d-separated) of Y_a , hence, including Z do not contribute any information for the model.

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GSMM and G-estimation

- The SMMs were introduced with a consistent estimation methodology called the G-estimation. Robins (1994) developed the G-estimation for the linear and log link functions.
- Vansteelandt and Goetghebeur (2002) later extended the theory by developing a G-estimator for the GSMM.
- The underlying idea of the G-estimation approach is to combine the structural and association models to construct a prediction of the counterfactual outcome for each subject. Then, this prediction and an the auxiliary function of the instrument Z is used to construct estimating equations.
- The values of the causal parameters that solve these estimating equations are called the G-estimators.
- Loosely speaking, the G-estimators are the values of the causal parameters ψ under which a valid IV assumption $Y_a \perp\!\!\!\perp Z | L$ holds in the observed data.

Special case of the G-estimation from the `ivtools` package

- Let $h(\psi)$ be the residual where the causal effect is removed from the factual outcome Y .
- Therefore, $h_i(\hat{\psi})$ as an estimator of the counter-factual value Y_0 , i.e., $\hat{Y}_{0i} = h_i(\hat{\psi})$.
- Since $Y_0 \perp\!\!\!\perp Z|L$ implies that IV assumptions are satisfied, we require $d(L, Z) \perp\!\!\!\perp Y_0|L$. Thus the value of ψ that solves the following estimating equation is the G-estimator of ψ

$$H(\psi) = \sum_{i=1}^n \hat{d}(L_i, Z_i) h_i(\psi) = 0,$$

where $d(L, Z)$ is arbitrary function such that $\mathbb{E}[d(L, Z)|L] = 0$ (more details below).

- The main idea: G-estimator is an estimator which makes the potential outcome Y_0 independent of the instrument Z in the sample, conditionally on the measured covariates L .

The $d(L, Z)$ function

The function $d(L, Z)$ is arbitrary function with the same dimension as ψ , such that $\mathbb{E}[d(L, Z)|L] = 0$. The `ivtools` package allows for two choices of $d(L, Z)$

1. $d(L, Z) = m(L)(Z - \mathbb{E}[Z|L])$
 2. $d(L, Z) = m(L)(\mathbb{E}[A|L, Z] - \mathbb{E}[A|L])$.
- The first function requires to model only the instrument-nuisance model $\mathbb{E}[Z|L]$, whereas the second function requires both $\mathbb{E}[A|L, Z]$ and exposure-nuisance model $\mathbb{E}[A|L]$.
 - The first function requires the instrument Z to be a scalar variable, whereas the second function allows for Z to be a vector of arbitrary length.

The link functions that `ivtools` offers; identity, log-link, logit

$$h(\psi) = \begin{cases} Y - m^T(L)A\psi \\ Y \exp\{-m^T(L)A\psi\} \\ \text{expit}\left(\text{logit}\left(\widehat{\mathbb{E}}[Y|L, Z, A]\right) - m^T(L)A\psi\right) \end{cases}$$

The general form of the $h(\psi)$ function

- Assume a double- ξ model. Namely, both the causal and association models are defined by the link function ξ . The GSMM is

$$\xi(\mathbb{E}[Y_1|L, Z, A = 1]) - \xi(\mathbb{E}[Y_0|L, Z, A = 1]) = m^T(L)A\psi,$$

and the association model of the observed outcome follows the same structure,

$$\xi(\mathbb{E}[Y|Z, L, A]).$$

- The general form of the $h(\psi)$ function is

$$h(\psi; L, Z, A) = \xi^{-1} \left(\xi(\mathbb{E}[Y|L, Z, A]) - m^T(L)A\psi \right).$$

Double-logit model example

- Assume a binary outcome $Y \in \{0, 1\}$, a binary exposure $A \in \{0, 1\}$, and a binary instrument $Z \in \{0, 1\}$, and assume that there are no measured confounders L , thus, $m(L) = 1$. For the logit link function ξ , the GSMM

$$\text{logit}(\mathbb{E}[Y_1|L, Z, A = 1]) - \text{logit}(\mathbb{E}[Y_0|L, Z, A = 1]) = \psi,$$

where $\text{logit}(x) = \ln\left(\frac{x}{1-x}\right)$. The causal parameter ψ represents the mean causal effect of the exposure on the exposed on the logit scale.

- The saturated association model of the observed outcome follows the same structure,

$$\text{logit}(\mathbb{E}[Y|Z, L, A = 1]) = \beta_0 + \beta_1 + \beta_2 Z + \beta_3 ZA.$$

- The vector of coefficients $\beta_Y = (\beta_0, \beta_1, \beta_2, \beta_3)^T$ incorporates the association between Z , A and Y which, at least partially, results from the omission of U .

Double-logit model example - continued

- Notably, if $U = \emptyset$, then $\beta_2 = \beta_3 = 0$ since conditioning on the exposure A blocks the information flow from the instrument Z to the outcome, i.e., $Y \perp\!\!\!\perp Z|L$; thus, $\beta_1 = \psi$.
- However, if there are unmeasured confounders U , then $\beta_1 \neq \psi$. Therefore, ψ cannot be estimated using the classical maximum likelihood method.
- However, the association model coefficients β_Y can still be estimated using the score equations that result from the maximum likelihood approach (although they do not convey causal meaning).
- In the logistic regression model $\xi^{-1}(x) = \text{expit}(x) = (1 + \exp\{-x\})^{-1}$, hence

$$\mathbb{E}[Y|L, Z, A] = P(Y = 1|L, Z, A) = \text{expit}(\psi) = (1 + \exp\{-\psi\})^{-1}.$$

- Namely, for the double-logit model $h(\psi)$ is

$$h(\psi; L, Z, A) = \text{expit}(\beta_0 + \beta_1 A + \beta_2 Z + \beta_3 ZL - \psi).$$

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Mendelian randomization

- The underlying principle of the method is to use genotypes as IVs to estimate the causal effect of phenotype on disease-related outcomes.
- The population distribution of genetic variants is assumed to be independent of behavioural and environmental factors that usually confound the effect of exposure on the outcome.
- The process governing the distribution of genetic variants in the population resembles the randomisation mechanism in RCTs.

Vitamin D data description

- The phenotype is vitamin D status, the outcome is survival at the follow-up endpoint, and the genotype is mutations in the filaggrin gene.
- These mutations are associated with a higher serum vitamin D concentration.
- We used the modified version of data obtained in the Monica10 population-based study. This is a 10-years follow-up study that started in 1982-1984 and initially included an examination of 3,785 individuals of Danish origin.

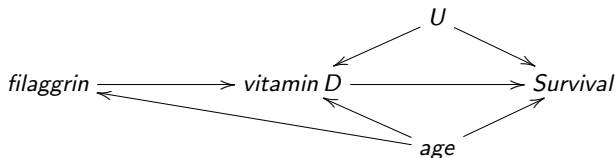
Data description

- We use a data set VitD borrowed from Martinussen et al. (2018) on a cohort study of vitamin D and mortality.
- We will estimate the causal effect of vitamin D on mortality by IV analysis, using mutations in the filaggrin gene as an instrument.
- These mutations have been shown to be associated with higher vitamin D status (Skaaby et al. 2013).
- The VitD data frame contains 2571 subjects and the following 5 variables:
 - 1 age (at baseline) - A measured covariate (confounder) L .
 - 2 filaggrin (binary indicator of mutations presence) - An instrument Z .
 - 3 vitd (vitamin D level) - The exposure A .
 - 4 time (follow-up time) - The time-to-event outcome T .
 - 5 death (a binary indicator of a subject's death) - The binary point outcome Y .

The vitamin D example

Data preparation

- We use the binary indicator of survival at the follow-up endpoint as the outcome Y .
- We used the binary indicator of the presence of the filaggrin gene mutation as the IV Z .
- We use the continuous vitamin D level as the exposure A .
- We use the continuous age variable as a measured confounder L .
- The estimated model was the double logit model.



Instrument-nuisance model; $\text{logit}(\mathbb{E}[Z|L]) = \theta_0 + \theta_1 L$

```
glm(formula = filaggrin ~ age, family = binomial(link = "logit"),  
data = VitD)
```

	Estimate	Std.Error	z value	Pr(> z)
(Intercept)	-2.920480	0.394538	-7.402	1.34e-13 ***
age	0.007447	0.006911	1.078	0.281

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1375.6 on 2570 degrees of freedom

Residual deviance: 1374.5 on 2569 degrees of freedom

AIC: 1378.5

The vitamin D data: G-estimation for binary outcome

Outcome model: $\text{logit}(\mathbb{E}[Y|L, Z, A]) = \beta_0 + \beta_1 L + \beta_2 Z + \beta_3 A$

```
glm(formula = death ~ age + filaggrin + vitd2,  
family = binomial(link = "logit"), data = VitD)
```

	Estimate	Std.Error	z value	Pr(> z)
(Intercept)	-7.876339	0.370140	-21.279	< 2e-16 ***
age	0.120590	0.005833	20.675	< 2e-16 ***
filaggrin	-0.358969	0.210633	-1.704	0.088337.
vitd2	-0.165277	0.043124	-3.833	0.000127 ***

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2803.2 on 2570 degrees of freedom

Residual deviance: 2196.4 on 2567 degrees of freedom

AIC: 2204.4

G-estimator of the causal effect ψ

```
ivglm(estmethod = "g", X = "vitd2", fitZ.L = fitZ.L2,  
fitY.LZX = fitY.LZX2, data = VitD, link = "logit")
```

	Estimate	Std.Error	z value	Pr(> z)
vitd2	-1.423	1.844	-0.772	0.44

- The G -estimator of the mean causal effect ψ is statistically insignificant.

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Data generating processes

We assume that $m(L) = 1$, an unmeasured confounder $U \sim N(10, 9)$, a continuous instrument $Z \sim N(3, 1)$, and a random error $\varepsilon \sim N(0, 1)$, thus we can define the exposure and the outcome processes

$$X = 1 + U + 2Z + 3\varepsilon, \quad Y = 1.5 + 3A + 2U + 3\varepsilon,$$

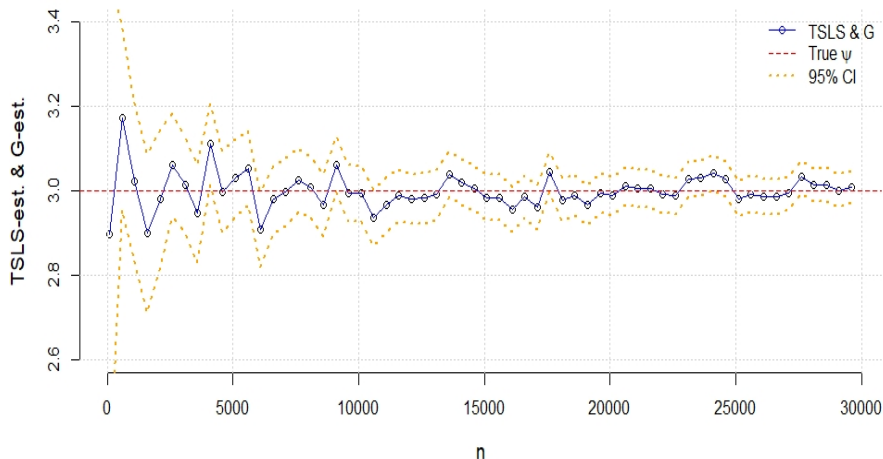
the true causal effect ψ is 3. If one omits U and estimates the model

$$Y = \beta_0 + \beta_1 A + \sigma\varepsilon,$$

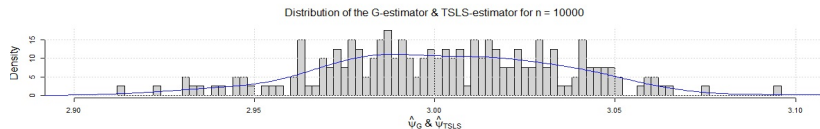
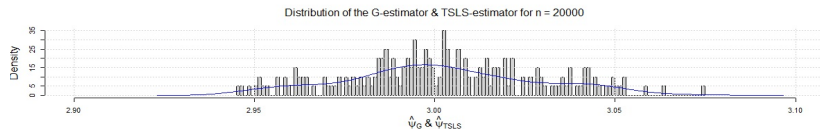
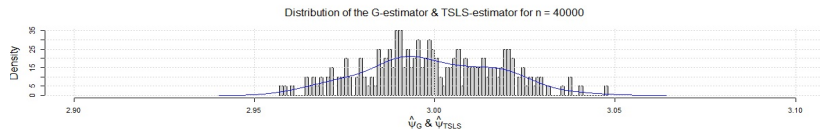
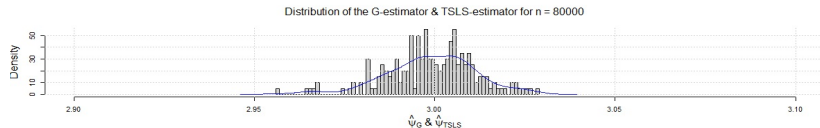
then the OLS of β_1 converges to $3.82 > \psi$. Namely, we need to apply either TSLS or G-estimation.

Consistency of the TSLS and G estimators in linear model

Consistency of the G-estimator and the TSLS-estimator $\hat{\psi}_{\text{TSLS}}$, $\hat{\psi}_{\text{G}}$ for linear link-function η



Large samples distribution of the estimator in linear model



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Data generating processes

We assume a measured confounder of A and Y that we wish to control for, $L \sim N(2, 9)$, however no causal interaction effect, i.e., $m(L) = 1$, and a binary instrument $Z \sim \text{Ber}(0.4)$, thus we can define a binary exposure process $X \sim \text{Ber}(p_x(L, Z))$, where

$$p_x(L, Z) \equiv \mathbb{E}[A|L, Z] \equiv \mathbb{P}(A = 1|L, Z) = \text{expit}(1 + 2Z + 0.3L),$$

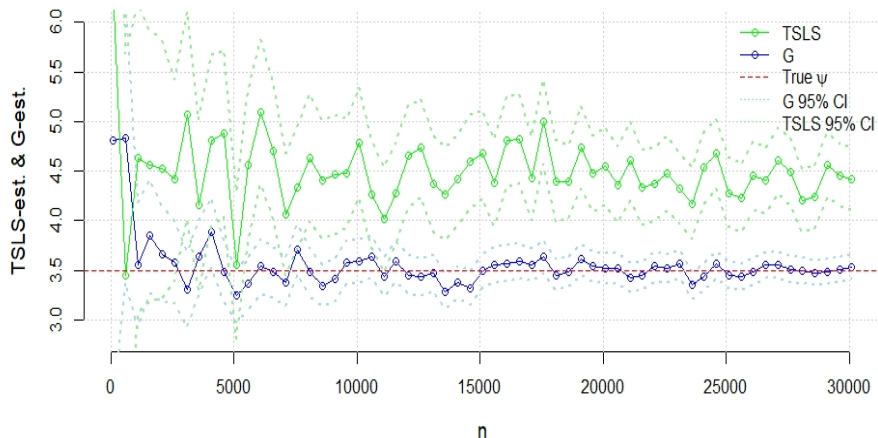
and the outcome process $Y \sim \text{Ber}(p_y(L, A))$, where

$$p_y(L, A) \equiv \mathbb{E}[Y|L, A, Z] \equiv \mathbb{P}(Y = 1|L, A) = \text{expit}(-1 + 3.5A - 0.5L),$$

such that $\text{expit}(x)$ is the logistic function, i.e., $\text{expit}(x) = (1 + e^{-x})^{-1}$ and the true causal effect ψ is set to be 3.5.

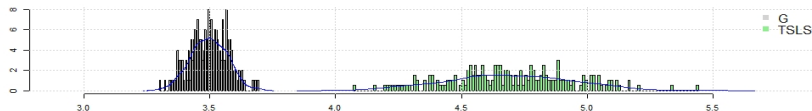
Consistency of a G estimator in logistic model

Consistency of the G-estimator $\hat{\psi}_G$ & inconsistency of the TSLS-estimator $\hat{\psi}_{\text{TSLS}}$ for logit link-function η

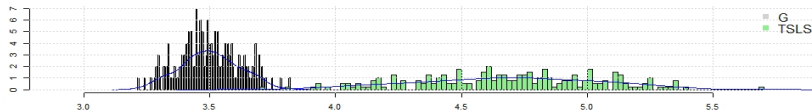


Large sample distributions of the TLSL and G estimators in logistic model

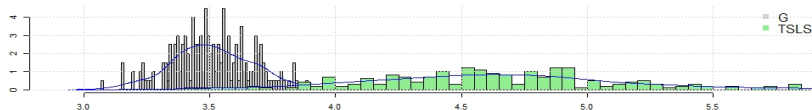
Distribution of the G and the TLSL estimators for $n = 8000$



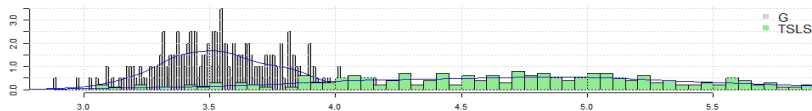
Distribution of the G and the TLSL estimators for $n = 4000$



Distribution of the G and the TLSL estimators for $n = 2000$



Distribution of the G and the TLSL estimators for $n = 1000$



Linear model example

- As we have shown analytically, for the linear link function, both estimators (TSLS and G) are identical, therefore:
- Both estimators are consistent and asymptotically normal.

Logistic model example

- The G-estimator is consistent and quite stable.
- The TSLS estimator is inconsistent and unstable.

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Concluding remarks on TSLS and G-estimation

- The IV analysis allows to mimic a randomised experiment.
- The TSLS requires two models: One for the exposure as a function of the instrument and another for the outcome as a function of the predicted exposure.
- The G-estimation requires one to two models. One for the instrument and another for the outcome as a function of the exposure and the instrument. The second model is required only for non-linear link functions.
- The two-stage estimator of the causal effect is consistent only for the linear (identity) link function.
- G-estimator allows us to estimate (consistently) the causal effect both in linear and non-linear models.
- For the linear link function, the two-stage estimator equals the G-estimator.
- G-estimator is an estimator which makes the potential outcome Y_0 independent of Z in the sample, conditionally on L .
- R-package `ivtools` enables the user to estimate the causal effect in models with linear, log, and logit link functions.
- R-package `ivtools` provides analytical standard errors of the causal effect estimator, which obviates the need for time-consuming bootstrap procedures.

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